

Aufgabe 1

Protokoll Ausgabe:

UDP endpoint is (192.168.2.42, 64590)

Received {"sequence":0,"ring":[{"ip":"192.168.2.42","port":64591}]} from 192.168.2.42:64591

Token: seq=0, #members=1 (192.168.2.42, 64591)

Sending {"sequence":1,"ring":[{"ip":"192.168.2.42","port":64591},{ip":"192.168.2.42","port":64590}]} to 192.168.2.42:64590

Received {"sequence":1,"ring":[{"ip":"192.168.2.42","port":64591},{ip":"192.168.2.42","port":64590}]} from 192.168.2.42:64590

Token: seq=1, #members=2 (192.168.2.42, 64591) (192.168.2.42, 64590)

Sending {"sequence":2,"ring":[{"ip":"192.168.2.42","port":64590},{ip":"192.168.2.42","port":64591}]} to 192.168.2.42:64591

Received {"sequence":3,"ring":[{"ip":"192.168.2.42","port":64591},{ip":"192.168.2.42","port":64590}]} from 192.168.2.42:64591

Token: seq=3, #members=2 (192.168.2.42, 64591) (192.168.2.42, 64590)

Sending {"sequence":4,"ring":[{"ip":"192.168.2.42","port":64590},{ip":"192.168.2.42","port":64591}]} to 192.168.2.42:64591

Received {"sequence":5,"ring":[{"ip":"192.168.2.42","port":64591},{ip":"192.168.2.42","port":64590}]} from 192.168.2.42:64591

Token: seq=5, #members=2 (192.168.2.42, 64591) (192.168.2.42, 64590)

Sending {"sequence":6,"ring":[{"ip":"192.168.2.42","port":64590},{ip":"192.168.2.42","port":64591}]} to 192.168.2.42:64591

Received {"sequence":7,"ring":[{"ip":"192.168.2.42","port":64591},{ip":"192.168.2.42","port":64590}]} from 192.168.2.42:64591

Token: seq=7, #members=2 (192.168.2.42, 64591) (192.168.2.42, 64590)

Sending {"sequence":8,"ring":[{"ip":"192.168.2.42","port":64590},{ip":"192.168.2.42","port":64591}]} to 192.168.2.42:64591

Wireshark Protokoll

0000	02 00 00 00	45 00 00 7b	c3 42 00 00	80 11 00 00	...E...{ -B- ...
0010	c0 a8 02 2a	c0 a8 02 2a	fc 4f fc 4e	00 67 8a 46	...*...* -O-N-g F
0020	7b 22 73 65	71 75 65 6e	63 65 22 3a	31 33 33 2c	{"sequen ce":133,
0030	22 72 69 6e	67 22 3a 5b	7b 22 69 70	22 3a 22 31	"ring":[{"ip":"1
0040	39 32 2e 31	36 38 2e 32	2e 34 32 22	2c 22 70 6f	92.168.2 .42","po
0050	72 74 22 3a	36 34 35 39	31 7d 2c 7b	22 69 70 22	rt":6459 1},{ "ip"
0060	3a 22 31 39	32 2e 31 36	38 2e 32 2e	34 32 22 2c	:"192.16 8.2.42",
0070	22 70 6f 72	74 22 3a 36	34 35 39 30	7d 5d 7d	"port":6 4590}]}

Aufgabe 2

Protokoll der Ausgabe:

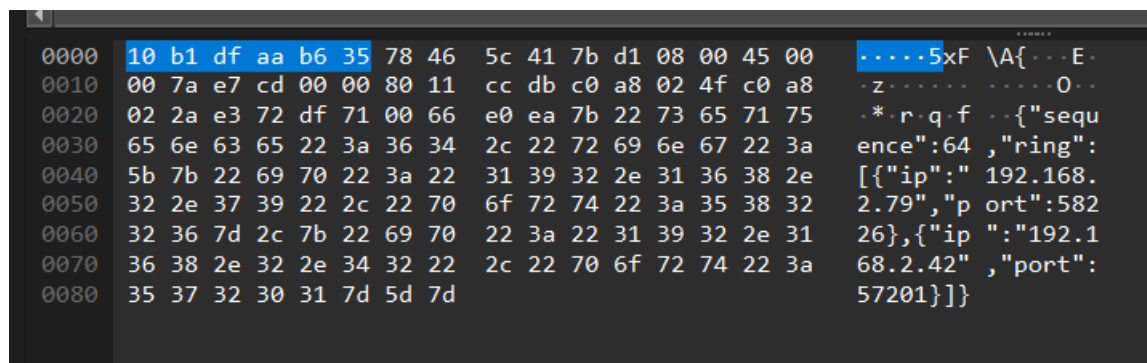
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Received {"sequence":0,"ring":[{"ip":"192.168.2.42","port":64220}]} from 192.168.2.42:64220
Token: seq=0, #members=1 (192.168.2.42, 64220)
Sending {"sequence":1,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} to
192.168.2.79:59430
Received {"sequence":1,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} from
192.168.2.79:59430
Token: seq=1, #members=2 (192.168.2.42, 64220) (192.168.2.79, 59430)
Sending {"sequence":2,"ring":[{"ip":"192.168.2.79","port":59430}, {"ip":"192.168.2.42","port":64220}]} to
192.168.2.42:64220
Received {"sequence":3,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} from
192.168.2.42:64220
Token: seq=3, #members=2 (192.168.2.42, 64220) (192.168.2.79, 59430)
Sending {"sequence":4,"ring":[{"ip":"192.168.2.79","port":59430}, {"ip":"192.168.2.42","port":64220}]} to
192.168.2.42:64220
Received {"sequence":5,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} from
192.168.2.42:64220
Token: seq=5, #members=2 (192.168.2.42, 64220) (192.168.2.79, 59430)
Sending {"sequence":6,"ring":[{"ip":"192.168.2.79","port":59430}, {"ip":"192.168.2.42","port":64220}]} to
192.168.2.42:64220
Received {"sequence":7,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} from
192.168.2.42:64220
Token: seq=7, #members=2 (192.168.2.42, 64220) (192.168.2.79, 59430)
Sending {"sequence":8,"ring":[{"ip":"192.168.2.79","port":59430}, {"ip":"192.168.2.42","port":64220}]} to
192.168.2.42:64220
Received {"sequence":9,"ring":[{"ip":"192.168.2.42","port":64220}, {"ip":"192.168.2.79","port":59430}]} from
192.168.2.42:64220
Token: seq=9, #members=2 (192.168.2.42, 64220) (192.168.2.79, 59430)

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Wireshark result

udp.port == 57201 or udp.port == 58226 als Filter



Aufgabe 4:

Probleme gab es erstmals bei der Ausführung einerseits wegen einer veralteten jdk andererseits, der Frage wie man das Programm zweimal startet. Beides konnte ich aber recht schnell lösen. Da ich keinen Kontakt zu den anderen Teilnehmern habe, habe ich jetzt nur meinen Laptop und Desktop als unterschiedliche PCs verwendet.